Promoting CO_2-Assisted Oxidative Dehydrogenation of Propane over attractive CeZn and CuCeZn catalysts

Cedric Karel Fonzeu Monguen [a, b], *Samuel Daniel* [b, c], *Hannington Nevin Otieno* [b, c], *Patrick Lott* [a], *Olaf Deutschmann* [a], *Zhen-Yu Tian* [b, c]

[a] *Institute for Chemical Technology and Polymer Chemistry, Karlsruhe Institute of Technology (KIT), Engesserstr. 20, Karlsruhe 76131, Germany*

[b] *Institute of Engineering Thermophysics, Chinese Academy of Sciences, Beijing 100190, China*

[c] *University of Chinese Academy of Sciences, Beijing 100049, China*

Email: cedric.monguen@kit.edu

Propene is a crucial raw material in the production of acrylic acid, polypropylene, acrolein, polyacrylonitrile, and other petrochemicals[1]. It is obtained through the naphtha steam-cracking process, fluid catalytic petroleum cracking, and direct dehydrogenation of propane (DHP)[2]. DHP as one of the pathways for propene production, is characterized by a high energy consumption due to its endothermic reaction process, which can lead to the formation of coke on the catalyst surface[3]. The utilization of the greenhouse gas CO_2 for ODHP can contribute to achieving net-zero carbon emissions[4]. Furthermore, the introduction of CO_2 can help to remove surface-adsorbed hydrogen, thereby enhancing both propane conversion and selectivity[4].

In this study, a Ce/Zn binary oxide and the CuCeZn ternary oxide were synthesized via the impregnation method and was used for CO_2-ODHP at 550 °C. Herein, the Cu loading of Cu and its incorporation into Ce/Zn lattice was investigated for CO_2-ODHP. In addition, a detailed insight into the structure-activity dualism and the relationship between the Cu, Ce, and Zn was explored using *operando* X-ray-absorption fine structure (EXAFS) and X-ray absorption near edge spectroscopy (XANES) analysis.

Results and Discussion

The results of CO_2-ODP reactions over CuCeZn catalysts are given in Fig. 1. While increasing the Cu-loading from 0 to 3 wt.% increases the initial conversion of C_3H_8 (Fig. 1a), CO_2 conversion decreases (Fig. 1b); the highest initial C_3H_6 selectivity is obtained over 3%Cu(Ce/Zn. The synergies between the Cu metal in the trimetallic

system [Cu (Ce/Zn)] was found to enhance the selectivity mainly due to higher activity of Cu metal in promoted system rather than the bimetallic (Ce/Zn) one.

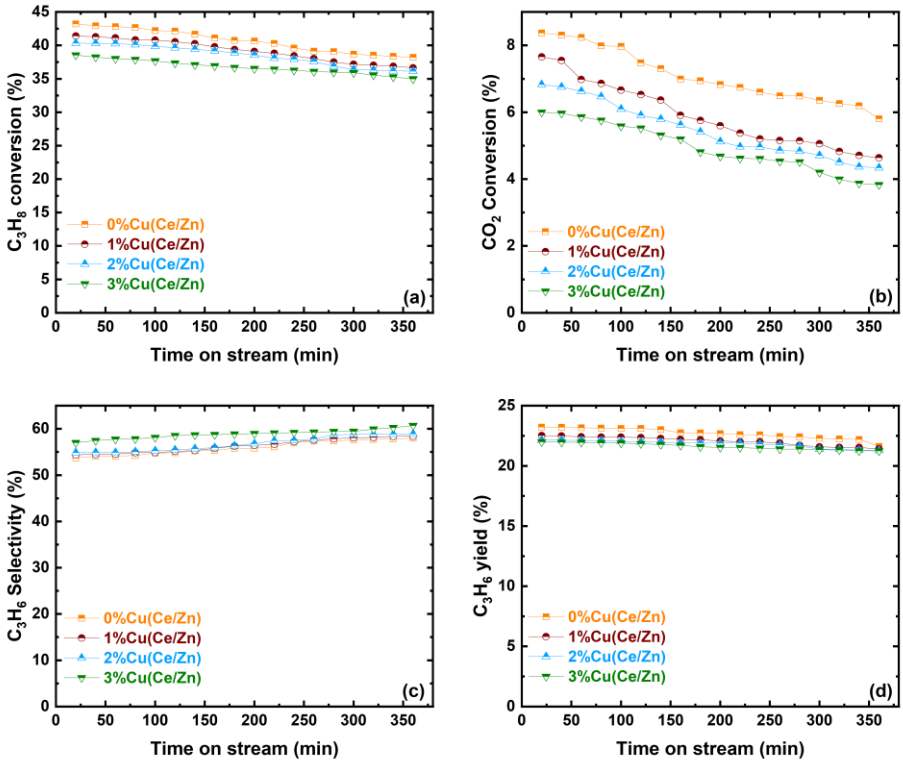

Figure 1. The time-on-stream conversions of C_3H_8 (a) and CO_2 (b), the C_3H_6 selectivity (c), and the C_3H_6 yield (d) for CO_2-ODPH catalyzed by CeZn and CuCeZn. GHSV = 12 000 h^{-1}, and gas composition: 4 % C_3H_8, 8 % CO_2, diluted in Ar. Total Flow Rate = 40 mL min^{-1}. T = 550 °C.

Significance

The CeZn binary oxide catalyst exhibited attractive performance for CO_2-ODHP. Loading Cu into the Ce/Zn structure was found to promote the propene selectivity with CO_2. The obtained results help to reveal the optimum loading of copper into (Ce/Zn) as an efficient catalyst for CO_2-ODHP. Thus, the novel catalyst materials proposed herein exhibit high potential for application in the petrochemical industry.

References

1. C.A. Carrero, R. Schloegl, I.E. Wachs, R. Schomaecker, ACS Catal. **2014**, 4, 3357–3380.

2. S. Chen, X. Chang, G. Sun, T. Zhang, Y. Xu, Y. Wang, C. Pei, J. Gong, Chem. Soc. Rev. **2021**, 50, 3315-3354.

3. J.H. Carter, T. Bere, J.R. Pitchers, D.G. Hewes, B.D. Vandegehuchte, C.J. Kiely, S.H. Taylor, G. Hutchings, Green Chem. **2021**, 23, 9747–9799.

4. S. Daniel, C. K. Fonzeu Monguen, Z.-Y. Tian, Chem. Eng. J. **2024**, 485, 150009.